

CFD Mesh Refinement Using Self-Organizing Maps

Evaluation of SOM-Based and Hybrid Refinement Indicators
for the Lid-Driven Cavity

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Abstract

This technical report investigates whether information learned by a Self-Organizing Map (SOM) can contribute to adaptive mesh-refinement decisions in computational fluid dynamics. A two-dimensional lid-driven cavity at Reynolds number $Re = 100$ is used as a controlled benchmark, with a fine uniform M160 solution serving as a numerical reference for post-solution evaluation.

The study first compares conventional velocity-gradient refinement with direct refinement based on SOM quantization error (SOM-QE) and with a hybrid gradient-SOM indicator under controlled refinement budgets. Direct SOM-QE refinement does not outperform the conventional Gradient criterion and exhibits non-monotonic behavior with increasing refinement budget. This negative result motivates a reformulation of the role of the SOM: rather than treating quantization error as a proxy for discretization error, the SOM is used to identify spatial transitions between SOM-classified flow states.

A SOM-interface indicator is subsequently constructed from BMU-state transitions, local velocity-gradient magnitude, and velocity jumps across neighboring SOM states. Using a fixed set of 40 priority seeds followed by one- and two-neighbor expansion, the resulting adaptive meshes are compared with Gradient meshes having matched final cell counts. For this cavity and the tested configuration, SOM-interface produces lower mean and RMS discrepancies relative to the M160 reference at both matched mesh sizes, while Gradient produces slightly lower maximum discrepancies. These results are therefore evidence for the tested configuration rather than a claim of general superiority.

A controlled sensitivity study further evaluates SOM architectures from 3×3 to 6×6 , with five random initializations per architecture. The density of SOM-state interfaces depends strongly on map resolution, while exact top-40 seed selection also exhibits initialization sensitivity. However, for the nominal 4×4 SOM, mean spatial agreement between seed sets increases from 66.5% for exact cell matching to 89.5% within one base-grid cell and 96.0% within two cells, indicating that much of the initialization sensitivity corresponds to local shifts rather than wholesale relocation of the selected regions.

The study demonstrates a reproducible workflow for separating SOM representation quality from numerical refinement utility and for evaluating machine-learning-derived mesh indicators without using the numerical reference during cell selection. The conclusions remain limited to a single laminar benchmark, one Reynolds number, and one-shot adaptive refinement; broader CFD cases and repeated adaptation cycles are required before assessing general applicability.

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1 Introduction

2 CFD Model

2.1 Benchmark problem

The two-dimensional lid-driven cavity was selected as the initial benchmark problem because it provides a simple geometry while producing a non-trivial recirculating incompressible flow field. The problem has also been extensively used for the verification and comparison of numerical methods. Classical reference solutions and detailed analyses of this benchmark are available in Ghia et al., Botella and Peyret, and Shankar and Deshpande [?, ?, ?].

The computational domain is a square cavity with characteristic length

$$L = 0.1 \text{ m.} \quad (1)$$

The upper wall moves in the positive x -direction with constant velocity

$$U_{\text{lid}} = 1 \text{ m s}^{-1}, \quad (2)$$

while the remaining cavity walls are stationary. The kinematic viscosity is

$$\nu = 1.0 \times 10^{-3} \text{ m}^2 \text{ s}^{-1}. \quad (3)$$

The corresponding Reynolds number is therefore

$$Re = \frac{U_{\text{lid}} L}{\nu} = 100. \quad (4)$$

Although the OpenFOAM computational mesh has a finite thickness in the z -direction, the front and back boundaries are defined using the `empty` boundary condition. The calculation therefore represents a two-dimensional flow problem.

2.2 Numerical solution

The incompressible transient Navier–Stokes equations are solved using OpenFOAM v2606. The `icoFoam` solver is used for the present benchmark. The numerical formulation follows the finite-volume framework underlying OpenFOAM; general finite-volume CFD methodology is described by Ferziger and Perić and by Versteeg and Malalasekera [?, ?, ?, ?].

The velocity field is denoted by $\mathbf{U}$, and the pressure field by p . The governing equations are

$$\nabla \cdot \mathbf{U} = 0, \quad (5)$$

and

$$\frac{\partial \mathbf{U}}{\partial t} + (\mathbf{U} \cdot \nabla) \mathbf{U} = -\nabla p + \nu \nabla^2 \mathbf{U}. \quad (6)$$

The CFD solution obtained on the initial mesh provides the physical information subsequently used to construct conventional and SOM-based refinement indicators.

The machine-learning model therefore does not replace the governing equations or the CFD solver. Instead, it operates on quantities extracted from a physics-based CFD solution.

3 Numerical Verification

Before evaluating machine-learning-based refinement strategies, a uniform-mesh refinement study was performed to establish the numerical behavior of the baseline CFD model. Grid-refinement studies and observed-order analyses are standard components of CFD solution verification [?, ?].

Four structured meshes were considered, denoted M20, M40, M80, and M160. Successive refinement approximately halves the characteristic cell spacing in each coordinate direction.

Table 1: Uniform meshes used for the numerical verification study.

Mesh	Cells	Δx [m]	Minimum U_x/U_{lid}
M20	400	0.00500	-0.198089
M40	1600	0.00250	-0.209591
M80	6400	0.00125	-0.212887
M160	25600	0.000625	-0.213733

A vertical velocity sampling line located at the horizontal center of the cavity,

$$x/L = 0.5, \quad (7)$$

was used to compare the horizontal velocity component between successive meshes. The resulting inter-grid differences are summarized in Table 2.

Table 2: Successive-mesh differences for the vertical centerline velocity profile.

Mesh comparison	L_2 difference	L_∞ difference
M20–M40	6.1826×10^{-3}	1.2598×10^{-2}
M40–M80	1.7211×10^{-3}	3.5090×10^{-3}
M80–M160	4.2841×10^{-4}	8.8600×10^{-4}

The reduction ratios of the successive L_2 differences are approximately

$$\frac{E_{20,40}}{E_{40,80}} = 3.59, \quad \frac{E_{40,80}}{E_{80,160}} = 4.02. \quad (8)$$

For a refinement ratio of approximately two, an observed convergence order can be estimated from the successive solution differences, consistent with standard grid-refinement verification practice [?, ?]:

$$p = \frac{\ln(E_h/E_{h/2})}{\ln(2)}. \quad (9)$$

The corresponding observed orders are approximately

$$p = 1.845 \quad (10)$$

for the M20–M40–M80 sequence and

$$p = 2.006 \quad (11)$$

for the M40–M80–M160 sequence.

These results indicate strong convergence of the sampled centerline velocity profile, with behavior approaching second order on the finer meshes.

3.1 Reference solution used in the adaptive-mesh study

The M160 solution is subsequently used as a fine-grid numerical reference for comparisons with the adaptive meshes. It is important to emphasize that M160 is not treated as an exact analytical solution or as a formally grid-independent solution.

Consequently, quantities reported later as “error” relative to M160 should more precisely be interpreted as discrepancies with respect to the selected fine-grid numerical reference.

3.2 Verification limitations

The time step was reduced together with the spatial mesh size during the uniform refinement sequence. Therefore, the present refinement study does not rigorously separate spatial discretization error from temporal discretization effects.

Furthermore, the discontinuity between the moving lid and the stationary side walls produces strong gradients near the upper corners of the cavity. These regions can dominate pointwise error measures and must be considered carefully when interpreting maximum-error metrics.

For these reasons, the verification study is used to establish numerical consistency and to define a controlled internal reference for the subsequent comparisons, rather than to claim an exact discretization-error estimate.

4 Self-Organizing Map Methodology

4.1 Purpose of the SOM

A Self-Organizing Map (SOM) was introduced to investigate whether unsupervised learning could identify physically meaningful flow states in the CFD solution and whether information derived from those states could subsequently contribute to mesh-refinement decisions.

The SOM is an unsupervised neural-network method that maps multidimensional input data onto a lower-dimensional discrete lattice while attempting to preserve topological relationships in the input space [?, ?]. SOM-based clustering and interpretation of the trained map have subsequently been developed for exploratory analysis of multidimensional data [?].

In the present work, the SOM is not used as a surrogate for the CFD solver. The Navier–Stokes equations remain the source of the physical solution. Instead, the SOM analyzes quantities extracted from the CFD field.

The initial SOM analysis is performed using the M40 solution, containing 1600 cells. Importantly, the M160 reference solution is not used during SOM training or during the construction of the refinement indicators. It is reserved for subsequent evaluation.

The supporting data-analysis workflow was implemented in Python using the scientific-computing ecosystem. Array and numerical operations relied on NumPy and SciPy, while preprocessing utilities were drawn from scikit-learn [?, ?, ?].

4.2 Candidate physical features

For each M40 cell, the CFD post-processing pipeline extracts the cell-center coordinates and local flow quantities. The candidate nondimensional quantities include

$$\frac{U_x}{U_{\text{lid}}}, \quad \frac{U_y}{U_{\text{lid}}}, \quad \frac{|\mathbf{U}|}{U_{\text{lid}}}, \quad \frac{L|\nabla\mathbf{U}|}{U_{\text{lid}}}, \quad \frac{L|\boldsymbol{\omega}|}{U_{\text{lid}}}, \quad (12)$$

where

$$\boldsymbol{\omega} = \nabla \times \mathbf{U}. \quad (13)$$

The velocity-gradient magnitude is computed using the Frobenius norm of the velocity-gradient tensor.

Spatial coordinates are deliberately excluded from the SOM input vector. Consequently, any spatially coherent classification produced by the SOM must arise from similarities in the local flow state rather than from explicit knowledge of cell position.

4.3 Feature selection

An exploratory correlation analysis showed a strong correlation between the velocity-gradient magnitude and the vorticity magnitude,

$$r(|\nabla \mathbf{U}|, |\boldsymbol{\omega}|) = 0.981. \quad (14)$$

To reduce this strong redundancy, vorticity was excluded from the first SOM input vector. The resulting feature vector was

$$\mathbf{x}_i = [U_{x,i}^* \quad U_{y,i}^* \quad |\mathbf{U}_i|^* \quad |\nabla \mathbf{U}_i|^*]^T, \quad (15)$$

where the superscript $*$ denotes physical nondimensionalization using the characteristic cavity length and lid velocity.

The velocity magnitude is mathematically related to its two velocity components and therefore introduces some redundancy. It was retained in this initial investigation to preserve an explicit scalar measure of local flow speed. Feature-ablation studies are left for future work.

4.4 Standardization

Physical nondimensionalization and machine-learning standardization are treated as separate operations.

After nondimensionalization, each SOM feature is standardized according to

$$z_j = \frac{x_j - \mu_j}{\sigma_j}, \quad (16)$$

where μ_j and σ_j are respectively the mean and standard deviation of feature j over the M40 dataset.

This prevents variables with larger numerical ranges from dominating the Euclidean distance used during SOM training.

4.5 SOM architecture and training

The SOM consists of a 4×4 rectangular lattice containing 16 neurons. The implementation was developed in Python using NumPy, with feature standardization performed using `StandardScaler`.

Training used 20,000 iterations with a fixed random seed of 42 to provide reproducibility. The learning rate decreased exponentially from

$$\alpha_0 = 0.5 \quad (17)$$

to

$$\alpha_f = 0.05, \quad (18)$$

while the neighborhood radius decreased from

$$\sigma_0 = 2.0 \quad (19)$$

to

$$\sigma_f = 0.5. \quad (20)$$

A Gaussian neighborhood function was used during weight updates.

For each CFD cell, the best-matching unit (BMU) is the neuron whose weight vector has the smallest Euclidean distance to the standardized feature vector.

4.6 Quantization error

The cell-level SOM quantization error is defined as

$$QE_i = \|\mathbf{z}_i - \mathbf{w}_{b(i)}\|_2, \quad (21)$$

where $\mathbf{z}_i$ is the standardized feature vector of cell i and $\mathbf{w}_{b(i)}$ is the weight vector of its BMU. For the trained M40 SOM, the mean cell-level quantization error was

$$\overline{QE} = 0.4976, \quad (22)$$

while the maximum value was

$$QE_{\max} = 7.5142. \quad (23)$$

All 16 neurons contained assigned CFD cells after training.

A large quantization error indicates that a CFD state is represented relatively poorly by its assigned SOM neuron. It does not, by definition, represent CFD discretization error. Distinguishing these two quantities is central to the experiments presented later in this report.

4.7 Interpretation of SOM states

Despite the exclusion of the spatial coordinates from the training features, the resulting BMU assignments form spatially coherent regions when mapped back onto the cavity.

This observation indicates that the SOM is identifying recurring local flow states rather than arbitrary spatial locations. Different neurons represent combinations of velocity direction, velocity magnitude, and velocity-gradient magnitude.

The SOM therefore provides a classification of the local physical state. Whether this classification contains information useful for adaptive refinement is treated as an experimental question rather than an assumption.

5 Adaptive Mesh Refinement Methodology

Adaptive mesh refinement is a well-established strategy for concentrating computational resolution in selected regions of a numerical domain. Classical AMR methods provide important historical context for the general concept of local refinement [?, ?]. The cell-selection indicators developed in the present work are not implementations of those classical AMR algorithms; they constitute a separate one-shot refinement experiment built around the OpenFOAM solution and the SOM-derived information.

5.1 Controlled comparison

Three cell-selection strategies were evaluated under identical refinement budgets:

1. a conventional velocity-gradient indicator;
2. direct refinement based on SOM quantization error (SOM-QE);

3. a hybrid indicator combining the velocity gradient with SOM-derived regime-transition information.

For each method, refinement budgets of 5%, 10%, 15%, and 20% of the 1600-cell M40 baseline mesh were considered.

The corresponding numbers of initially selected cells were therefore 80, 160, 240, and 320 cells, respectively.

The same local refinement operation was applied after cell selection. Consequently, the final adaptive meshes contained 1840, 2080, 2320, and 2560 cells for the four refinement budgets.

This design isolates the effect of the cell-selection criterion: at a given budget, each method starts from the same M40 solution, selects the same number of cells, and uses the same refinement procedure.

5.2 Velocity-gradient indicator

The conventional baseline ranks cells according to the local velocity-gradient magnitude,

$$G_i = |\nabla \mathbf{U}_i|. \quad (24)$$

Cells with the largest values of G_i are selected until the prescribed refinement budget is reached.

This method provides a physically interpretable conventional baseline: regions in which the velocity field varies rapidly receive higher refinement priority.

5.3 Direct SOM quantization-error indicator

The first SOM-based hypothesis uses the cell-level quantization error directly as a refinement indicator,

$$I_{QE,i} = QE_i. \quad (25)$$

Cells with the largest quantization errors are selected first.

This criterion tests whether CFD states that are poorly represented by the trained SOM also correspond to locations where additional spatial resolution is beneficial.

No fine-grid reference information is used during this selection.

5.4 SOM regime-transition indicator

The BMU classification provides a discrete label b_i for every M40 cell. To identify interfaces between different SOM-classified flow states, a local transition indicator is defined using the direct structured-grid neighbors of cell i :

$$T_i = \frac{\sum_{j \in \mathcal{N}_i} \mathbb{I}(b_j \neq b_i)}{|\mathcal{N}_i|}, \quad (26)$$

where $\mathcal{N}_i$ is the set of existing direct neighbors and $\mathbb{I}$ is the indicator function.

Thus,

$$0 \leq T_i \leq 1. \quad (27)$$

A value of zero indicates that all direct neighbors belong to the same SOM state, while larger values indicate increasing local interaction between different SOM-classified states.

The transition indicator is constructed exclusively from the M40 SOM classification. It does not use the M160 reference solution.

The cell-level correlation between the transition indicator and the velocity-gradient magnitude was approximately

$$r(G, T) = 0.261, \quad (28)$$

substantially smaller than the correlation between gradient magnitude and quantization error,

$$r(G, QE) = 0.879. \quad (29)$$

The transition indicator therefore contains information that is less redundant with the conventional gradient criterion than the direct quantization error.

5.5 Hybrid-v1 indicator

The hybrid indicator was designed to preserve the conventional gradient as the primary refinement signal while allowing SOM regime-transition information to modify the ranking of cells.

The normalized gradient is defined as

$$G_i^* = \frac{G_i}{\max_j G_j}. \quad (30)$$

The Hybrid-v1 indicator is then

$$I_{\text{hyb},i} = G_i^* (1 + \alpha T_i), \quad (31)$$

with

$$\alpha = 0.5. \quad (32)$$

The coefficient was fixed for the Hybrid-v1 experiment and was not optimized against the M160 reference.

This distinction is important because tuning the coefficient to minimize M160 discrepancies and subsequently evaluating performance against the same M160 solution would contaminate the validation comparison.

Hybrid-v1 should therefore be interpreted as a predefined exploratory indicator rather than an optimized model.

5.6 Selection overlap

Because the gradient remains the dominant component of Hybrid-v1, the hybrid criterion modifies only a relatively small fraction of the conventional gradient selections.

The overlap between Gradient and Hybrid-v1 selections was

Table 3: Overlap between Gradient and Hybrid-v1 cell selections.

Budget	Selected	Common	Exclusive per method	Overlap
5%	80	73	7	91.25%
10%	160	149	11	93.12%
15%	240	227	13	94.58%
20%	320	306	14	95.62%

Consequently, Hybrid-v1 is not a replacement of the gradient criterion by a machine-learning criterion. It is a controlled perturbation of the gradient-based ranking using information extracted from SOM flow-state boundaries.

5.7 Second-stage SOM-interface refinement criterion

The results obtained with direct SOM quantization error motivated a change in the evaluation question. Rather than treating the SOM as a direct estimator of discretization error, the subsequent analysis asked whether the SOM classification contains information about *where refinement is likely to be useful*.

The BMU field provides a spatial classification of local flow states. Cells located at interfaces between different BMU states were therefore considered potentially informative because they represent locations where the local flow regime changes in the SOM feature space.

For an interface cell, the local transition fraction T_i was combined with two quantities already available from the M40 pilot solution. The first was the normalized velocity-gradient magnitude,

$$G_i^* = \frac{G_i}{\max_j G_j}, \quad (33)$$

and the second was the normalized maximum velocity-vector jump across a direct SOM interface,

$$\Delta U_i^* = \frac{\max_{j \in \mathcal{N}_i, b_j \neq b_i} \|\mathbf{U}_i - \mathbf{U}_j\|_2}{\max_k \Delta U_k}. \quad (34)$$

The resulting SOM-interface ranking was defined as

$$I_{\text{SI},i} = T_i \sqrt{G_i^* \Delta U_i^*}. \quad (35)$$

Only cells for which $T_i > 0$ were eligible for interface selection. The highest-ranked 40 interface cells were then frozen as common seeds. Starting from exactly those same seeds, the selected refinement region was expanded by a fixed graph distance: one face-sharing layer for the $1N$ case and two layers for the $2N$ case.

This construction deliberately separates two questions. The SOM determines *where flow-state transitions occur*, while the gradient and velocity-jump terms determine how strongly those transitions are ranked. Neither the M160 solution nor any reference-derived error field is used in this construction.

The resulting experiments selected 78 cells for $1N$ and 114 cells for $2N$. These counts were not forced to equal a nominal 10% budget. Instead, they were subsequently reproduced exactly with conventional Gradient selection, creating cell-matched controls with identical final cell counts. This comparison isolates the spatial-selection mechanism while avoiding a confounding difference in total mesh size.

5.8 Evaluation against the fine-grid reference

After refinement, each adaptive case is solved independently with OpenFOAM. The resulting velocity field is compared with the M160 fine-grid numerical reference.

Because the adaptive and reference meshes contain different cell-center locations, the M160 solution is interpolated to the adaptive-mesh cell centers before the velocity discrepancy is evaluated.

For cell i , the nondimensional velocity-vector discrepancy is

$$e_i = \frac{\|\mathbf{U}_{i,\text{adaptive}} - \mathbf{U}_{i,\text{M160}}\|_2}{U_{\text{lid}}}. \quad (36)$$

Since adaptive cells have different volumes, global mean and RMS measures are volume weighted.

The M160 solution is used only at this evaluation stage and not during the selection of cells for Gradient, SOM-QE, or Hybrid-v1.

6 Results

6.1 Controlled adaptive-refinement matrix

The three refinement strategies were evaluated at four identical refinement budgets, producing a controlled matrix of 12 adaptive CFD simulations. Table 4 summarizes the principal results.

Table 4: Adaptive-mesh results relative to the M160 fine-grid numerical reference. Mean and RMS velocity discrepancies are volume weighted. CPU time corresponds to the final CFD solve and does not include the complete adaptive workflow cost.

Method	Budget	Cells	CPU [s]	Mean error	RMS error
Gradient	5%	1840	9.64	0.001846	0.003994
Gradient	10%	2080	11.64	0.001656	0.003756
Gradient	15%	2320	13.62	0.001296	0.003505
Gradient	20%	2560	14.05	0.001105	0.003384
SOM-QE	5%	1840	9.62	0.002349	0.004392
SOM-QE	10%	2080	11.24	0.002628	0.004492
SOM-QE	15%	2320	12.53	0.002291	0.004175
SOM-QE	20%	2560	13.77	0.002005	0.003924
Hybrid-v1	5%	1840	9.32	0.001841	0.004031
Hybrid-v1	10%	2080	11.09	0.001583	0.003709
Hybrid-v1	15%	2320	12.50	0.001394	0.003556
Hybrid-v1	20%	2560	14.68	0.001242	0.003453

All adaptive meshes passed the mesh-quality checks used in the automated experiment framework. The maximum Courant number remained below the imposed limit of 1.05 for all cases.

6.2 RMS discrepancy

Figure 1 compares the volume-weighted RMS velocity discrepancy as a function of final adaptive-mesh cell count.

The Gradient criterion exhibits monotonic improvement as the refinement budget increases. Its RMS discrepancy decreases from approximately 3.994×10^{-3} at the 5% budget to 3.384×10^{-3} at the 20% budget.

Direct SOM-QE refinement does not exhibit the same behavior. In particular, increasing the budget from 5% to 10% increases the RMS discrepancy from

$$4.392 \times 10^{-3} \tag{37}$$

to

$$4.492 \times 10^{-3}. \tag{38}$$

The SOM-QE criterion subsequently improves at the 15% and 20% budgets, but its RMS discrepancy remains higher than that of Gradient at every tested budget.

At equal refinement budgets, the Gradient RMS discrepancy is lower than the SOM-QE result by approximately 9.1%, 16.4%, 16.0%, and 13.7% for the 5%, 10%, 15%, and 20% budgets, respectively.

These results do not support the hypothesis that SOM quantization error can be used directly as a superior refinement indicator for the present cavity problem.

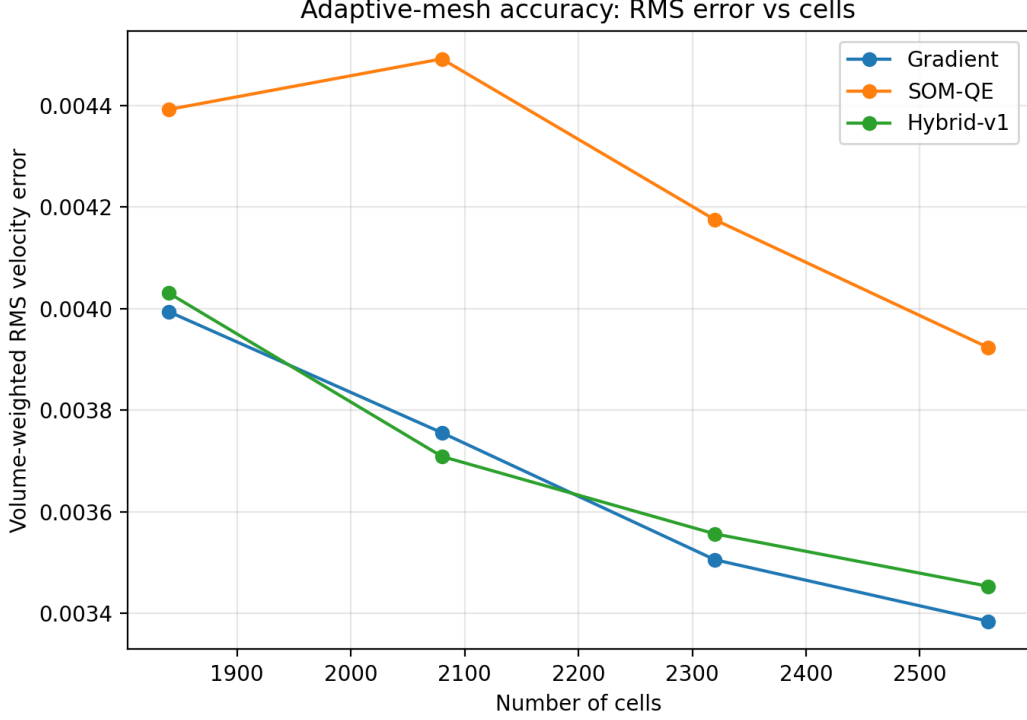


Figure 1: Volume-weighted RMS velocity discrepancy relative to the M160 fine-grid numerical reference for Gradient, SOM-QE, and Hybrid-v1 refinement.

6.3 Hybrid-v1 performance

Hybrid-v1 recovers monotonic RMS improvement with increasing refinement budget and remains close to the Gradient baseline.

Relative to Gradient, the Hybrid-v1 RMS discrepancy changes by approximately

$$-0.92\%, \quad +1.24\%, \quad -1.45\%, \quad -2.03\% \quad (39)$$

at the 5%, 10%, 15%, and 20% budgets, respectively, where a positive value denotes an improvement relative to Gradient.

Thus, Hybrid-v1 produces a small RMS improvement at the 10% budget, but this improvement is not reproduced at the other tested budgets. The present results therefore do not demonstrate a systematic accuracy advantage of Hybrid-v1 over the conventional Gradient criterion.

6.4 Mean velocity discrepancy

Figure 2 presents the corresponding volume-weighted mean velocity discrepancy.

The mean-error results reinforce the RMS comparison. Gradient improves monotonically from approximately 1.846×10^{-3} to 1.105×10^{-3} .

SOM-QE again exhibits non-monotonic behavior between the 5% and 10% budgets. At the four tested budgets, the Gradient mean discrepancy is approximately 21.4%, 37.0%, 43.4%, and 44.9% lower than the corresponding SOM-QE result.

Hybrid-v1 produces a mean-error improvement of approximately 0.25% and 4.43% relative to Gradient at the 5% and 10% budgets, respectively. However, at 15% and 20% its mean discrepancy is approximately 7.59% and 12.44% higher than Gradient.

As with the RMS metric, the hybrid modification therefore does not provide a consistent advantage across refinement budgets.

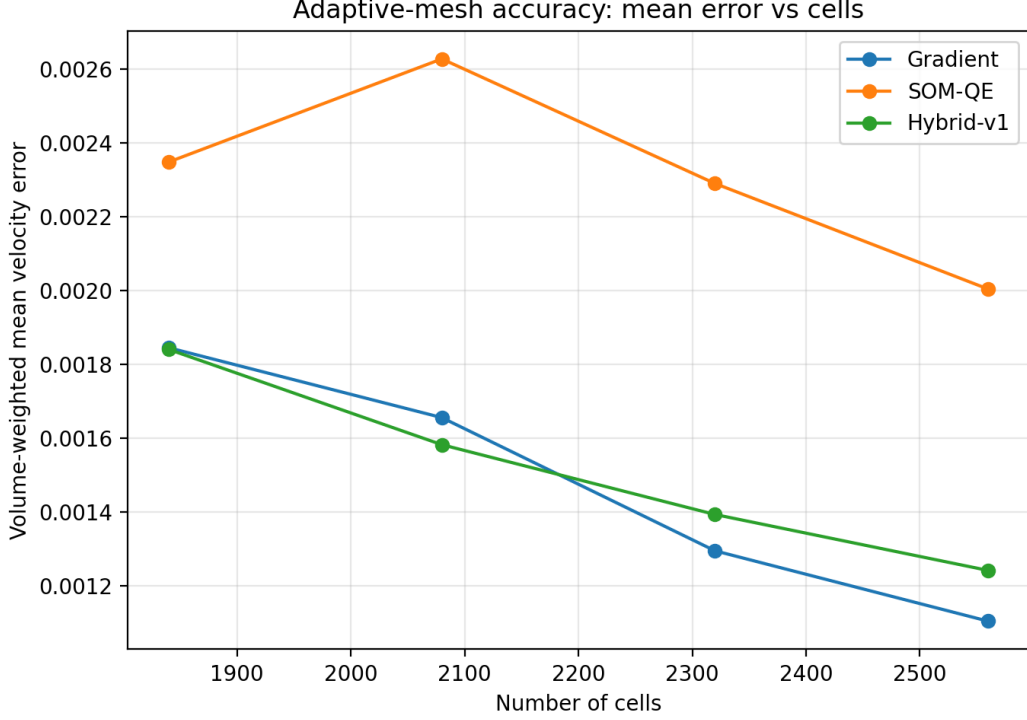


Figure 2: Volume-weighted mean velocity discrepancy relative to the M160 fine-grid numerical reference for the three refinement strategies.

6.5 What Hybrid-v1 changes

Because Hybrid-v1 preserves the gradient as its primary signal, its cell selection remains strongly overlapping with the conventional Gradient selection.

Figure 3 shows the spatial distribution of common and exclusive selections at all four budgets.

The overlap increases from 91.25% at the 5% budget to 95.62% at the 20% budget. Hybrid-v1 therefore acts as a controlled modification of the Gradient ranking rather than as an independent replacement criterion.

To identify the physical nature of these changes, the cells selected exclusively by each method were analyzed using only quantities available from the M40 pilot solution. Table 5 summarizes the results.

Table 5: Mean characteristics of cells selected exclusively by Gradient or Hybrid-v1. The analysis uses only M40 quantities and does not use the M160 reference.

Budget	Exclusive selection	Cells	Mean $ \nabla U ^*$	Mean T
5%	Gradient	7	7.489	0.000
5%	Hybrid-v1	7	6.663	0.619
10%	Gradient	11	5.384	0.000
10%	Hybrid-v1	11	4.810	0.583
15%	Gradient	13	3.730	0.000
15%	Hybrid-v1	13	3.261	0.596
20%	Gradient	14	2.836	0.054
20%	Hybrid-v1	14	2.461	0.643

A consistent pattern is observed at all four budgets. Hybrid-v1 removes a small number of cells having somewhat larger velocity gradients but little or no SOM regime-transition activity and replaces them with cells having slightly smaller gradients but substantially larger transition

Gradient vs Hybrid-v1 cell selection

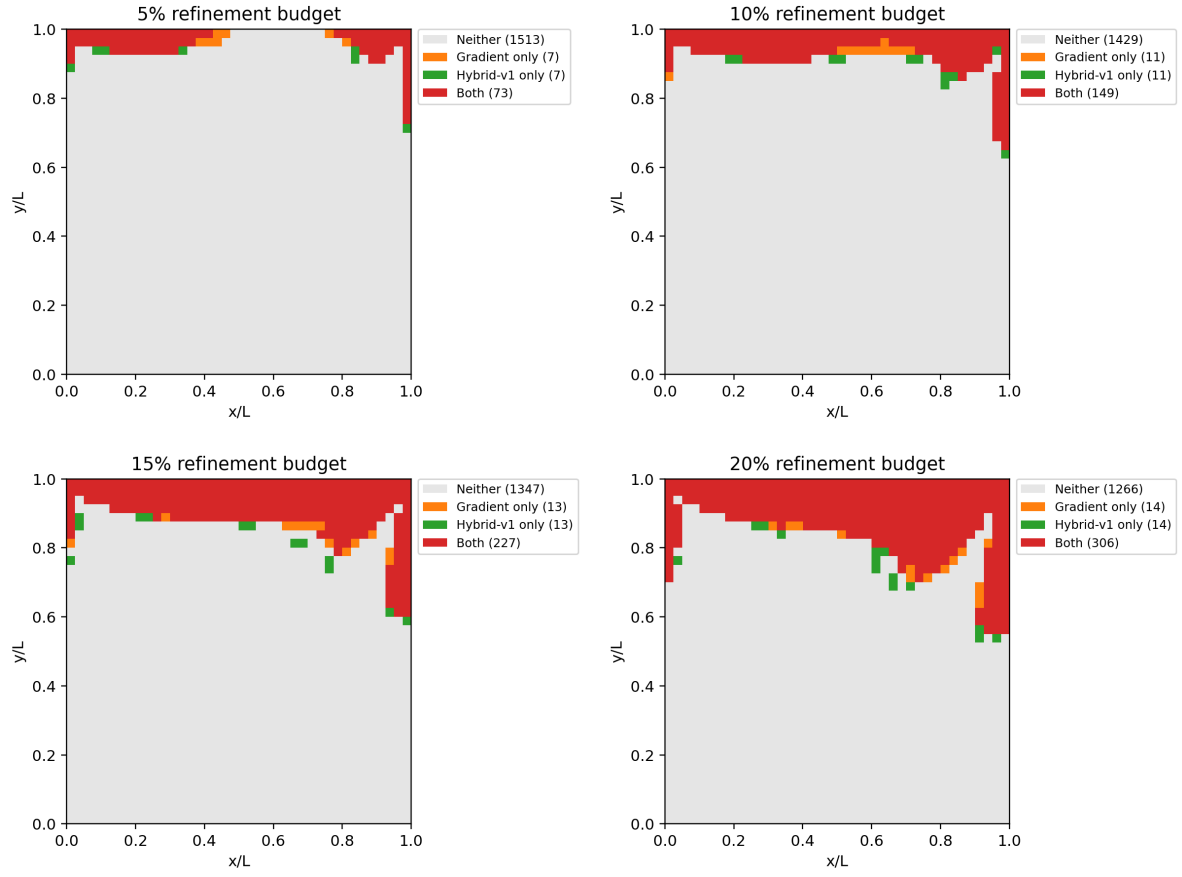


Figure 3: Comparison of Gradient and Hybrid-v1 cell selections for refinement budgets of 5%, 10%, 15%, and 20%. The hybrid formulation changes only a small subset of cells close to the selection threshold.

indicators.

For example, at the 10% budget the Gradient-only cells have mean values

$$|\nabla U|^* = 5.384, \quad T = 0, \quad (40)$$

whereas the Hybrid-only cells have

$$|\nabla U|^* = 4.810, \quad T = 0.583. \quad (41)$$

The hybrid formulation is therefore behaving according to its intended design: SOM information changes the priority of cells located near boundaries between SOM-classified flow regimes.

However, the adaptive CFD results demonstrate that this physically interpretable modification is not systematically beneficial. The 10% case improves the RMS discrepancy, whereas the same mechanism slightly worsens the RMS result at 5%, 15%, and 20%.

This distinction motivates a central result of the present investigation: information that identifies a physically distinct flow state or a boundary between states is not necessarily equivalent to information that predicts the numerical utility of refining that location.

6.6 Computational-time observations

The measured solver execution times increase with final cell count, as expected. Differences between methods at equal cell count are relatively small compared with the differences in accuracy.

No claim of computational-speed superiority is made from these measurements. Only one timing measurement was collected for each case, and the reported solver times exclude the cost of pilot CFD calculation, SOM training, indicator construction, mesh refinement, field mapping, and other workflow operations.

A rigorous computational-efficiency comparison would require repeated timing measurements and accounting for the complete adaptive workflow.

6.7 Second-stage SOM-interface experiments

The preceding results motivated a change in the SOM evaluation criterion. Direct quantization error was not treated as a discretization-error estimator. Instead, the SOM classification was used to identify interfaces between distinct local flow states, and these interfaces were ranked using the indicator

$$I_{SI} = T\sqrt{G^*\Delta U^*}. \quad (42)$$

A fixed set of 40 highest-ranked interface seeds was used for both experiments. The refinement region was then expanded by one graph layer (1N) or two graph layers (2N). The two cases selected 78 and 114 M40 parent cells, respectively, producing final meshes with 1834 and 1942 cells.

Table 6: SOM-interface experiments and cell-matched Gradient controls. The Gradient cases use exactly the same number of selected M40 parent cells and therefore produce the same final cell counts.

Case	Method	Seeds	Selected	Cells	Mean error	RMS error
1N	SOM-interface	40	78	1834	0.001570	0.003929
1N	Gradient	–	78	1834	0.001824	0.003986
2N	SOM-interface	40	114	1942	0.001359	0.003611
2N	Gradient	–	114	1942	0.001672	0.003761

All four cell-matched experiments completed successfully to $t = 5$. The maximum Courant numbers were 0.926 for SOM-interface $1N$, 0.930 for SOM-interface $2N$, 0.952 for Gradient-78, and 0.961 for Gradient-114, remaining below the imposed limit of 1.05.

At 1834 cells, the SOM-interface case reduced the volume-weighted mean velocity discrepancy by approximately 13.9% relative to the cell-matched Gradient case. The RMS discrepancy was reduced by approximately 1.4%.

At 1942 cells, the corresponding reductions were approximately 18.7% for the mean discrepancy and 4.0% for the RMS discrepancy.

The maximum pointwise discrepancy remained very similar for the paired cases and occurred at the same normalized upper-corner location. For the $1N$ pair, the SOM-interface maximum was 0.136204, compared with 0.136135 for Gradient. For the $2N$ pair, the corresponding values were 0.136151 and 0.136135. Thus, the observed improvement is primarily reflected in the volume-weighted global metrics rather than in the maximum pointwise discrepancy.

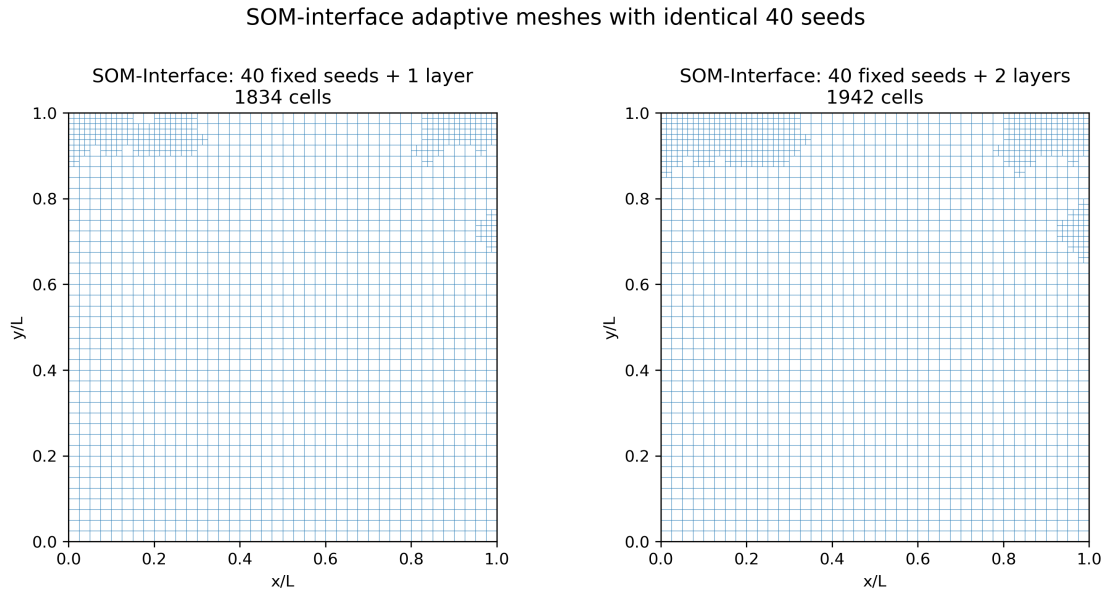


Figure 4: Final adaptive meshes produced by the SOM-interface criterion using the same 40 frozen interface seeds and one or two graph layers. The cases contain 1834 and 1942 cells, respectively.

The real-mesh comparison shows that the second-stage criterion does not simply refine isolated high-gradient cells. Instead, the fixed interface seeds generate spatially connected refinement regions. Increasing the graph radius from one to two layers preserves the same seed set and adds 36 additional cells, increasing the final mesh from 1834 to 1942 cells.

6.8 Gradient versus SOM-interface selection overlap

The cell-matched comparisons also show that the new criterion is related to, but not identical with, the conventional Gradient criterion. For the 1834-cell comparison, 53 of the 78 selected cells were common to both methods, corresponding to an overlap of 67.95%. Each method selected 25 additional cells not selected by the other.

For the 1942-cell comparison, 74 of the 114 selected cells were common, corresponding to an overlap of 64.91%. Gradient selected 40 cells that were not selected by SOM-interface, while SOM-interface selected another 40 cells.

These overlaps are important because the new results cannot be explained by a completely different refinement philosophy. A substantial portion of the selected cells is common to both

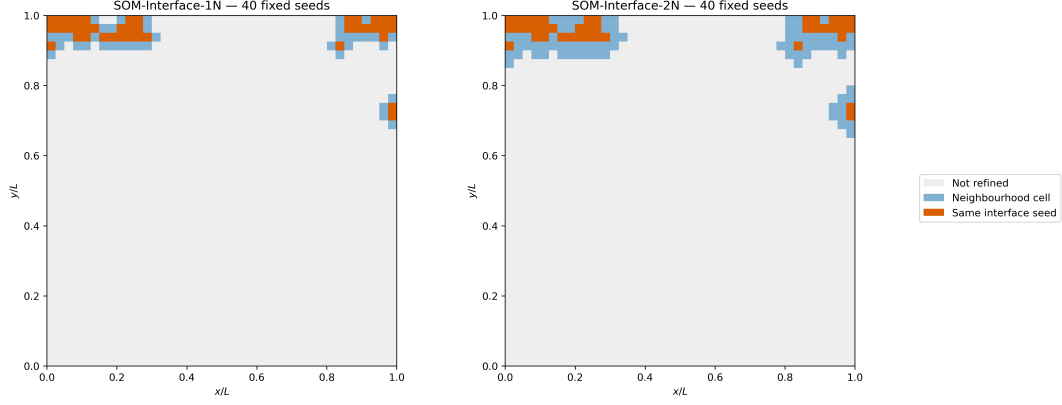


Figure 5: Selection obtained from the same 40 frozen SOM-interface seeds for one and two graph-layer expansions. The 1N selection is a subset of the 2N selection; the latter adds 36 cells.

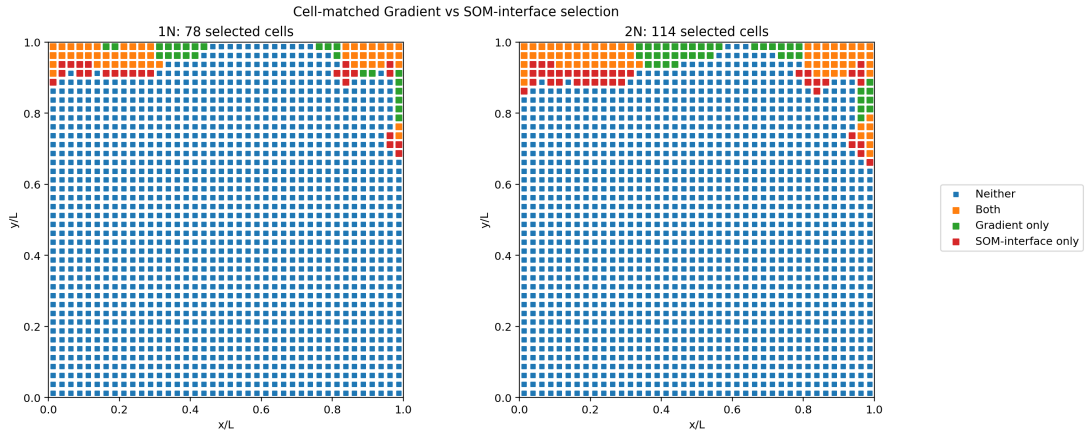


Figure 6: Cell-by-cell comparison between Gradient and SOM-interface selections at matched final mesh sizes. The methods share a substantial fraction of selected cells while allocating the remaining refinement budget differently.

methods. The numerical differences arise from the spatial allocation of the remaining refinement cells.

6.9 Refinement-utility analysis

To investigate whether the observed differences could be related to the utility of the selected regions, the adaptive-mesh discrepancy fields were interpolated back onto the original M40 cell centers. The M40–M160 discrepancy was used only as an evaluation quantity. Cells were partitioned into four mutually exclusive groups: selected by both methods, Gradient-only, SOM-only, and selected by neither method.

For the $1N$ comparison, the mean M40 discrepancy in the common region was 2.91×10^{-2} . The corresponding post-refinement errors were 7.43×10^{-3} for Gradient and 1.05×10^{-2} for SOM-interface. In the Gradient-only region, the post-refinement errors were 3.60×10^{-3} and 6.31×10^{-3} , respectively. In the SOM-only region they were 4.55×10^{-3} and 4.91×10^{-3} .

For the $2N$ comparison, the common region had a mean M40 discrepancy of 2.34×10^{-2} , with post-refinement errors of 6.44×10^{-3} for Gradient and 7.25×10^{-3} for SOM-interface. In the SOM-only region, however, SOM-interface produced a lower mean post-refinement error than Gradient, 3.50×10^{-3} versus 3.84×10^{-3} .

The same analysis also shows that the benefit of either refinement strategy is not confined to the cells explicitly selected for refinement. In the regions selected by neither method, SOM-interface produced lower mean post-refinement error than Gradient in both matched cases. This is consistent with the globally coupled nature of the CFD solution and reinforces the need to evaluate refinement strategies through the resulting solution rather than through local indicator values alone.

Table 7: Spatial refinement-utility analysis for the cell-matched comparisons. “SOM better fraction” is the fraction of M40 locations in each region for which the interpolated SOM-interface discrepancy is lower than the corresponding Gradient discrepancy.

Case	Region	N	M40 error	Gradient error	SOM error
1N	Both	53	2.907e-2	7.434e-3	1.053e-2
1N	Gradient-only	25	1.058e-2	3.603e-3	6.306e-3
1N	SOM-only	25	1.032e-2	4.546e-3	4.910e-3
1N	Neither	1497	2.205e-3	1.523e-3	1.145e-3
2N	Both	74	2.341e-2	6.445e-3	7.245e-3
2N	Gradient-only	40	5.936e-3	2.551e-3	2.940e-3
2N	SOM-only	40	9.194e-3	3.844e-3	3.496e-3
2N	Neither	1446	2.093e-3	1.332e-3	9.861e-4

The utility analysis should not be interpreted as a cell-level proof that SOM-interface is universally superior. Rather, it provides evidence that the new criterion changes the spatial distribution of refinement in a way that can produce a different global solution response. In particular, the $2N$ case shows a lower error in the SOM-only region, while the common and Gradient-only regions still favor Gradient on average.

6.10 Final adaptive-mesh topology

The differences between the refinement indicators can also be examined directly through the final computational meshes. Figure 7 compares the SOM-QE and Gradient meshes at the 10% refinement budget. Both simulations contain exactly 2080 cells and passed the OpenFOAM mesh quality checks.

The two indicators allocate the identical refinement budget differently. Gradient concentrates refinement predominantly near the moving lid and the upper cavity, where large velocity gradients

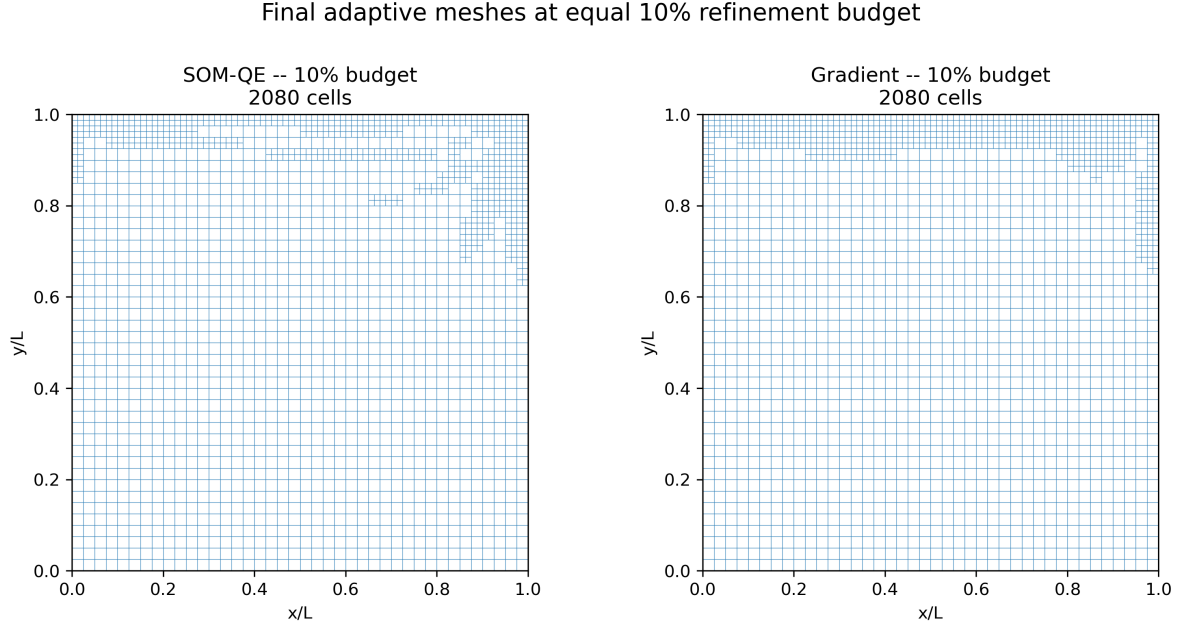


Figure 7: Final adaptive meshes obtained with SOM-QE and Gradient refinement at the same 10% refinement budget. Both meshes contain 2080 cells. The comparison therefore isolates differences in the spatial allocation of refinement rather than differences in total cell count.

occur. SOM-QE also refines the upper region but distributes part of its refinement budget into additional flow regions extending farther into the cavity.

This distinction is important because the adaptive-mesh comparison shows that equal numbers of cells do not imply equal numerical performance. At the 10% budget, the Gradient mesh produces a lower RMS discrepancy relative to the M160 reference despite having exactly the same number of cells as the SOM-QE mesh. The result therefore points to the spatial allocation of refinement, rather than cell count alone, as the relevant difference between the two strategies.

7 Discussion

7.1 Why direct SOM-QE refinement did not outperform Gradient

The controlled adaptive-mesh experiments provide a clear result for the present lid-driven-cavity problem: direct refinement based on SOM quantization error does not outperform the conventional velocity-gradient indicator.

This result is particularly relevant because the preliminary indicator analysis initially suggested that SOM-QE contained potentially useful information. On the M40 pilot solution, quantization error was spatially correlated with regions exhibiting relatively large discrepancies with the M160 reference. At some refinement budgets, ranking cells by SOM-QE captured a substantial fraction of the initial M40–M160 discrepancy.

However, the adaptive simulations demonstrate that identifying cells with large current discrepancy is not equivalent to identifying cells whose refinement will produce the largest reduction in the final global error.

The distinction may be written conceptually as

$$\text{current local discrepancy} \neq \text{refinement utility}. \quad (43)$$

Refinement modifies the discrete CFD system itself. The resulting velocity field is globally coupled through the governing equations, pressure–velocity interaction, boundary conditions,

and neighboring control volumes. Consequently, refining one region can improve the solution in locations that were not directly selected for refinement.

This behavior was observed explicitly in the 10% refinement-benefit analysis. Gradient refinement improved the solution over a broad fraction of the common diagnostic grid, including regions not directly selected by the Gradient criterion. SOM-QE refinement produced a smaller global benefit and even increased the local discrepancy in portions of the unselected domain.

Therefore, the value of a refinement indicator cannot be established solely from its correlation with an existing error field. It must ultimately be evaluated through the solution obtained after refinement.

7.2 What the SOM actually learned

The inferior performance of direct SOM-QE refinement should not be interpreted as evidence that the SOM failed to learn meaningful information about the flow.

The SOM was trained using local flow variables and velocity-gradient information without including the spatial coordinates x/L and y/L as input features. Nevertheless, the resulting best-matching-unit classification formed spatially coherent regions when mapped back onto the cavity.

This indicates that the SOM identified recurring local flow states rather than merely memorizing spatial position.

Analysis of the SOM neurons further showed that cells having comparable velocity-gradient magnitudes could belong to different neurons because their velocity magnitude and directional components differed. In this sense, the Gradient indicator and the SOM encode different types of information:

$$\text{Gradient} \rightarrow \text{strength of local solution variation}, \quad (44)$$

whereas

$$\text{SOM state} \rightarrow \text{type of local flow regime}. \quad (45)$$

This distinction explains why SOM-based information can be physically structured without automatically becoming an effective numerical-error indicator.

7.3 Quantization error is not an error estimator

SOM quantization error measures the distance between an input sample and its best-matching SOM prototype in standardized feature space. A large value therefore indicates that the local flow state is represented relatively poorly by the trained map.

It does not, by definition, measure discretization error.

In the present case, SOM-QE and velocity-gradient magnitude were strongly correlated. Their cell-level correlation was approximately

$$r(|\nabla U|, QE) = 0.879. \quad (46)$$

Both quantities therefore respond strongly to extreme flow states, including the upper-corner regions associated with the moving-lid boundary condition.

This explains why SOM-QE initially appeared promising as an error-location indicator. Nevertheless, the adaptive experiments separate correlation from causation: the cells with the largest quantization error are not necessarily the cells whose refinement most efficiently reduces the global CFD discrepancy.

For the present experiment, the hypothesis that SOM quantization error can serve directly as a superior refinement indicator is therefore not supported.

7.4 Interpretation of the final adaptive meshes

The final 10% adaptive meshes provide a geometric interpretation of the quantitative results.

Both SOM-QE and Gradient begin from the same M40 pilot mesh, select the same number of cells for refinement, and produce final meshes containing 2080 cells. Nevertheless, Figure 7 shows that the cells are distributed differently.

The Gradient criterion concentrates a larger fraction of the available resolution close to the moving lid and upper cavity, where the velocity field changes rapidly. SOM-QE allocates part of the same finite refinement budget to additional regions farther from the lid.

Those additional SOM-selected regions are not random. They correspond to flow states that the SOM regards as unusual or poorly represented. The adaptive CFD results nevertheless show that resolving those states more finely does not produce as much global numerical benefit as concentrating the same computational resources according to the Gradient criterion.

This provides an important engineering distinction:

$$\text{physically interesting} \neq \text{numerically profitable to refine.} \quad (47)$$

For adaptive CFD, the objective is not merely to identify interesting flow structures, but to allocate a limited computational budget where additional resolution most improves the quantities of interest.

7.5 What was learned from Hybrid-v1

The Hybrid-v1 indicator was deliberately designed after rejecting direct SOM-QE refinement as the primary strategy. Instead of allowing SOM information to replace the conventional indicator, the velocity gradient remained dominant and the SOM regime-transition indicator only modified its priority:

$$I_{\text{hyb},i} = G_i^* (1 + \alpha T_i), \quad \alpha = 0.5. \quad (48)$$

The coefficient was fixed before evaluating the hybrid cases against the M160 reference. It was not optimized using the reference solution.

This formulation produced a controlled perturbation of the Gradient selection. Depending on refinement budget, approximately 91–96% of the Gradient-selected cells were retained.

The cells introduced by Hybrid-v1 consistently had somewhat smaller velocity gradients but substantially larger SOM regime-transition indicators than the cells they replaced. The algorithm therefore behaved according to its intended design.

At the 10% budget, changing only 11 of the 160 Gradient-selected cells produced an approximately 1.24% reduction in RMS discrepancy. However, the same mechanism did not improve RMS performance at the 5%, 15%, or 20% budgets.

The appropriate conclusion is therefore not that the hybrid method improves Gradient refinement. Rather, the experiment demonstrates that SOM-derived flow-state information can alter refinement decisions in a controlled and physically interpretable way, while the numerical value of those changes remains dependent on the specific cells exchanged.

7.6 Refinement utility as the central problem

The combined results suggest a more precise research question for future work.

The original question can be formulated as whether a SOM can identify cells that should be refined. The present results indicate that this formulation is too broad. A more useful question is:

What information learned by a SOM, if any, is predictive of the numerical utility of refining a cell or region?

This distinction changes the role of machine learning in the adaptive workflow.

Rather than asking an unsupervised model to replace established numerical indicators, a more defensible strategy is to use conventional physics-based or numerical indicators as the primary refinement signal and investigate whether learned flow-state information can improve decisions near the refinement threshold.

The Hybrid-v1 experiment represents a first controlled test of this concept, but the present evidence is insufficient to establish a general advantage.

7.7 Reframing the SOM evaluation

The direct SOM-QE results changed the research question rather than ending the investigation. Quantization error was shown not to be a reliable direct proxy for discretization error or refinement utility. The subsequent analysis therefore used the SOM in a different role: identifying interfaces between distinct learned flow states.

This distinction is important. A SOM does not need to predict numerical error directly for its representation of the flow to be useful. It may instead identify spatial structures that complement a conventional numerical indicator. The SOM-interface criterion was constructed around this hypothesis.

The new indicator combines three pieces of information available from the M40 pilot solution:

$$I_{SI} = T\sqrt{G^*\Delta U^*}. \quad (49)$$

The transition term identifies a change in learned flow regime, while the gradient and velocity-jump terms retain sensitivity to the local strength of the physical variation. The resulting refinement is therefore not a pure machine-learning replacement of Gradient. It is a physics-informed use of the SOM representation.

7.8 Cell-matched evaluation of SOM-interface

A direct comparison at a nominal 10% budget would have been misleading for the new criterion because fixed-seed graph expansion naturally produces a different number of selected cells. The experiments were therefore followed by cell-matched Gradient controls.

The $1N$ SOM-interface case selected 78 cells and was compared with a Gradient selection of exactly 78 cells. The $2N$ case selected 114 cells and was compared with exactly 114 Gradient-selected cells. In both cases, the final meshes therefore contained identical numbers of cells.

Under this controlled comparison, SOM-interface produced lower volume-weighted mean and RMS discrepancies than the corresponding Gradient case. The mean discrepancy reduction was approximately 13.9% for the $1N$ comparison and 18.7% for the $2N$ comparison. The RMS reduction was smaller, approximately 1.4% and 4.0%, respectively.

The result is noteworthy because the methods shared a substantial fraction of their selected cells, yet allocated the remaining cells differently. The overlap decreased from 67.95% in the $1N$ comparison to 64.91% in the $2N$ comparison.

The observed improvement should nevertheless be interpreted carefully. The maximum point-wise discrepancy remained essentially unchanged between the paired cases, and the simulations represent only one Reynolds number and one cavity geometry. The present evidence therefore supports the statement that the SOM-interface hypothesis is promising for this controlled test, not that the method has been established as a generally superior adaptive strategy.

7.9 Why the new result is different from SOM-QE

The contrast between SOM-QE and SOM-interface is itself an important result. SOM-QE ranks cells according to how poorly their local feature vector is represented by a SOM prototype. SOM-interface instead uses the spatial organization of the BMU classification.

Consequently,

$$QE \rightarrow \text{representation mismatch}, \quad (50)$$

whereas

$$T \rightarrow \text{learned flow-state transition}. \quad (51)$$

The second-stage criterion further combines the transition with local physical variation. The new result therefore does not contradict the earlier negative SOM-QE experiment. It tests a different hypothesis about what information in the SOM may be useful for mesh adaptation.

7.10 Interpretation of the refinement-utility analysis

The cell-matched utility analysis provides a more nuanced interpretation than simply comparing global mean and RMS values. In the common selected region, Gradient produced lower mean post-refinement error in both the $1N$ and $2N$ comparisons. Gradient also produced lower mean error in the Gradient-only region.

In contrast, the SOM-interface selection showed lower mean error in the SOM-only region for the $2N$ case, and the SOM-interface solution also showed lower mean error in the large region selected by neither method. These observations indicate that the global improvement cannot be reduced to a statement that every SOM-selected cell is individually more useful than every Gradient-selected cell.

The result is consistent with the global coupling of the CFD equations. Refinement changes the solution throughout the computational domain, so the numerical value of a refinement decision must ultimately be evaluated through its downstream effect on the complete solution.

The utility analysis therefore supports a more cautious interpretation: the SOM-interface criterion appears to identify a different set of spatial structures whose refinement can alter the global solution beneficially, but the causal relationship between individual selected cells and global error reduction remains unresolved.

7.11 Engineering implications

The study illustrates a broader methodological point for machine-learning applications in computational engineering.

A machine-learning indicator can exhibit spatial structure, correlate with a reference discrepancy, and identify physically meaningful flow regimes while still failing to improve the final numerical solution.

For this reason, evaluation should proceed through a hierarchy of evidence:

1. verify the underlying CFD solution;
2. determine what information the machine-learning model actually represents;
3. compare the proposed indicator against a conventional baseline at equal computational budgets;
4. solve the adapted CFD cases rather than evaluating indicator correlation alone;
5. assess the final quantities of interest and global error metrics;
6. account for the complete computational cost before making efficiency claims.

Within the original four-budget SOM-QE/Hybrid-v1 matrix, the conventional Gradient indicator remains the most consistently accurate baseline. The subsequent SOM-interface experiments provide a different and more promising result in cell-matched tests, but they are not yet sufficient to establish general superiority. The principal contribution of the combined analysis is therefore the development and testing of a more precise hypothesis: learned flow-regime information may be useful for adaptive refinement when it is combined with physically meaningful measures of local variation and evaluated through the resulting CFD solution.

8 Limitations

The present study is intended as a controlled computational investigation rather than as a demonstration of a generally applicable adaptive-meshing method. Several limitations must therefore be considered when interpreting the results.

8.1 Fine-grid numerical reference

The M160 solution is used throughout the adaptive-mesh analysis as a fine-grid numerical reference. It should not be interpreted as an exact solution.

Consequently, quantities reported as errors relative to M160 are more precisely numerical discrepancies with respect to the finest solution used in this study. Although the uniform-mesh sequence showed strong convergence toward M160, a remaining discretization error is necessarily present in the reference itself.

The results therefore support comparisons between refinement strategies under a common reference, but they do not establish absolute discretization errors.

8.2 Spatial and temporal discretization

The uniform-grid convergence study changed the time step together with the spatial resolution. The reported M20–M40, M40–M80, and M80–M160 differences therefore do not isolate spatial discretization error independently from temporal discretization effects.

The observed convergence rates are consistent with strong numerical convergence of the tested sequence, particularly on the finer meshes, but they should not be interpreted as a rigorous spatial order-of-accuracy verification.

A future verification study should independently refine the time step and the spatial mesh.

8.3 Lid-driven-cavity corner behavior

The moving lid meets stationary side walls at the upper corners of the cavity. This discontinuity in the imposed wall velocity generates strong local gradients and influences pointwise discrepancy measures.

The largest M40–M160 velocity discrepancy occurs close to the upper-right corner, and a substantial fraction of the squared discrepancy is concentrated in a relatively small number of cells near the upper boundary.

This behavior can influence both conventional gradient indicators and SOM-derived quantities. In particular, the observed correlation between quantization error and the M40–M160 discrepancy is partly associated with both quantities responding strongly to extreme upper-corner flow states.

For this reason, maximum pointwise discrepancy is not used as the primary criterion for comparing adaptive strategies. Volume-weighted mean and RMS metrics provide more representative global measures for the present study.

8.4 Single flow configuration

All adaptive experiments were performed for the two-dimensional lid-driven-cavity problem at

$$Re = 100. \quad (52)$$

The conclusions therefore apply only to the tested configuration.

No evidence is presently available to determine whether the relative behavior of Gradient, SOM-QE, and Hybrid-v1 persists at different Reynolds numbers, for different geometries, in three-dimensional flows, or for problems involving heat transfer, turbulence, compressibility, multiphase phenomena, or other physical processes.

Generalization must therefore be tested rather than assumed.

8.5 SOM architecture and feature selection

The SOM configuration used in this investigation represents one specific model design: a 4×4 map containing 16 neurons and trained using the selected velocity and velocity-gradient features.

The present results do not establish that this architecture is optimal.

Similarly, the failure of direct SOM-QE refinement should not be interpreted as a general failure of self-organizing maps for adaptive CFD. It establishes only that quantization error from the present SOM configuration did not provide a superior direct refinement criterion for the tested problem.

Alternative features, SOM dimensions, training procedures, or derived indicators may produce different behavior. Such alternatives should, however, be defined without tuning against the numerical reference used for final evaluation.

8.6 Hybrid-v1 is not optimized

Hybrid-v1 uses

$$I_{\text{hyb}} = G^*(1 + 0.5T). \quad (53)$$

The coefficient $\alpha = 0.5$ was intentionally fixed rather than optimized against M160. This prevents the reference solution from being used both to design and evaluate the indicator.

Consequently, Hybrid-v1 should be interpreted as a proof-of-concept test of whether SOM regime-transition information can modify a conventional indicator in a controlled manner. It is not presented as an optimized adaptive-meshing algorithm.

The isolated improvement observed at the 10% budget is insufficient to establish superiority over Gradient.

8.7 Refinement-budget range

Only four refinement budgets were investigated:

$$5\%, \quad 10\%, \quad 15\%, \quad 20\%. \quad (54)$$

These experiments are sufficient to reveal important differences in monotonicity and relative behavior, but they do not fully characterize the accuracy–cost response of each method.

In particular, the present results should not be extrapolated to very small or very large refinement fractions without additional simulations.

8.8 Computational-cost measurements

The reported execution times correspond to individual CFD solves and were obtained from single runs. They are therefore susceptible to normal variability in workstation load and operating-system scheduling.

Furthermore, the timings do not include the complete adaptive workflow, which may contain:

- the pilot CFD calculation;
- feature extraction;
- SOM training;
- indicator calculation;
- cell selection;
- mesh refinement;
- field transfer and post-processing.

No claim of computational-speed superiority is therefore made.

A rigorous efficiency study should use repeated timing measurements and compare total workflow cost for a specified accuracy or quantity-of-interest target.

8.9 Refinement strategy

The current framework performs one refinement operation based on information from a pilot solution. It does not implement repeated adaptive cycles in which the solution is recomputed, the indicator reevaluated, and the mesh adapted iteratively.

The results therefore evaluate one-shot refinement decisions rather than a fully dynamic adaptive-mesh algorithm.

Multiple adaptation cycles could alter the relative behavior of the tested indicators and constitute a separate research question.

8.10 Cell-matched second-stage experiments

The SOM-interface experiments were deliberately evaluated against cell-matched Gradient controls because graph expansion produces selected-cell counts that are not equal to the nominal percentage used in the original adaptive matrix. The $1N$ and $2N$ cases therefore should not be combined directly with the original 5–20% budget curves as if they were additional points on the same budget sequence.

The cell-matched design controls final mesh size, but it does not eliminate all possible confounding factors. The SOM-interface method uses fixed seeds and graph expansion, whereas Gradient directly ranks individual cells. Differences in spatial connectivity and refinement topology are therefore part of the method being tested.

8.11 Scope of the conclusions

The strongest conclusion supported by the present evidence is deliberately limited.

For the tested $Re = 100$ lid-driven cavity, the conventional velocity-gradient indicator produced the most consistently accurate solutions in the original 5–20% SOM-QE/Hybrid-v1 refinement matrix. Direct SOM quantization-error refinement was inferior and non-monotonic, while Hybrid-v1 did not demonstrate a systematic advantage.

A subsequent, separately formulated SOM-interface criterion produced lower volume-weighted mean and RMS discrepancies than cell-matched Gradient controls at 1834 and 1942 cells. These

experiments are encouraging but remain limited to one geometry, one Reynolds number, one SOM configuration, and two graph-expansion choices. They therefore do not establish general superiority.

The present evidence supports the narrower conclusion that SOM-derived flow-state interfaces can provide information that is complementary to velocity-gradient magnitude. Whether this information can be converted into a robust and general predictor of refinement utility remains unresolved.

9 Conclusions and Future Work

9.1 Conclusions

This study investigated whether information extracted by a Self-Organizing Map (SOM) could provide useful guidance for adaptive mesh refinement in CFD. The two-dimensional lid-driven cavity at $Re = 100$ was used as a controlled test problem, and all proposed refinement strategies were compared at identical refinement budgets.

Before introducing machine learning, a uniform-mesh convergence study was performed using M20, M40, M80, and M160 meshes. The successive centerline differences decreased substantially with refinement, with observed convergence rates approaching second-order behavior on the finer mesh levels. The M160 solution was subsequently used as a fine-grid numerical reference rather than as an exact solution.

A 4×4 SOM was trained using flow information extracted from the M40 pilot solution. Spatial coordinates were deliberately excluded from the SOM input. Nevertheless, the trained map produced spatially coherent flow-state regions when its classifications were projected back onto the physical cavity. This demonstrates that the SOM learned structured relationships among the local flow variables rather than simply encoding spatial position.

SOM quantization error initially appeared potentially useful because it was concentrated in regions that also exhibited relatively large M40–M160 discrepancies. However, the controlled adaptive CFD experiments showed that this correspondence was insufficient to make quantization error an effective direct refinement indicator.

At all four tested refinement budgets, the conventional velocity-gradient criterion produced a lower RMS discrepancy than direct SOM-QE refinement. Gradient refinement also improved monotonically with increasing budget, whereas SOM-QE exhibited non-monotonic behavior between the 5% and 10% budgets.

The final adaptive meshes demonstrate that this difference cannot be attributed simply to the number of cells. At the 10% budget, both SOM-QE and Gradient produce meshes containing 2080 cells, but they allocate those cells differently. The Gradient strategy concentrates more resolution near regions of strong velocity variation, while SOM-QE allocates part of the same finite refinement budget to additional flow states that are unusual in SOM feature space.

The results therefore establish an important distinction:

$$\text{representation novelty} \neq \text{discretization error} \neq \text{refinement utility}. \quad (55)$$

A cell can be unusual from the perspective of an unsupervised machine-learning model without being the location where additional spatial resolution provides the greatest numerical benefit.

The refinement-benefit analysis reinforces this conclusion. Refining cells according to the Gradient criterion improved the solution over regions extending beyond the directly selected cells, illustrating the global coupling of the CFD solution. Consequently, local indicator quality cannot be assessed solely through correlation with a pre-existing local discrepancy field.

A second SOM-derived quantity, the regime-transition indicator T , was subsequently investigated. Unlike quantization error, this quantity was only weakly correlated with the velocity-gradient magnitude and therefore represented more distinct information.

This motivated the Hybrid-v1 indicator,

$$I_{\text{hyb}} = G^*(1 + 0.5T), \quad (56)$$

in which the conventional Gradient criterion remained dominant and SOM information only modified the ranking of cells near the refinement threshold.

Hybrid-v1 behaved as intended. Approximately 91–96% of the Gradient-selected cells were preserved, while a small number of high-transition cells replaced cells having somewhat larger gradients but little SOM-transition activity.

At the 10% budget, this modification reduced the RMS discrepancy by approximately 1.24% relative to Gradient. However, the improvement was not reproduced at the 5%, 15%, or 20% budgets.

Hybrid-v1 therefore does not demonstrate a systematic advantage over Gradient refinement.

The principal result of this investigation is consequently not that machine learning improves CFD mesh refinement. Instead, the study demonstrates that SOM-derived flow-state information can be physically structured and complementary to a conventional numerical indicator, while also showing that converting such information into reliable refinement decisions is a substantially more difficult problem.

For the present $Re = 100$ cavity experiments, the velocity-gradient criterion remains the most consistently accurate refinement strategy.

9.2 Second-stage SOM-interface result

The negative SOM-QE result led to a reformulation of the machine-learning hypothesis. Rather than treating quantization error as a direct error indicator, the SOM BMU field was used to identify interfaces between different learned flow states. A new ranking,

$$I_{\text{SI}} = T\sqrt{G^*\Delta U^*}, \quad (57)$$

combined the SOM transition fraction with normalized velocity-gradient magnitude and the local velocity-vector jump across the interface.

The criterion was evaluated using 40 fixed interface seeds and one- or two-layer graph expansion. The resulting selections contained 78 and 114 cells. Conventional Gradient controls were then constructed with exactly the same selected-cell counts, producing final meshes of 1834 and 1942 cells.

In these cell-matched comparisons, SOM-interface reduced the volume-weighted mean velocity discrepancy by approximately 13.9% and 18.7%, and the RMS discrepancy by approximately 1.4% and 4.0%, for the $1N$ and $2N$ cases, respectively.

These results represent a change in the evidence compared with the earlier SOM-QE experiments. They do not show that SOM information is intrinsically superior to gradient-based refinement. Instead, they show that the *type* of SOM information used matters: direct representation error performed poorly, whereas spatial transitions between learned flow states, combined with local physical variation, produced a different and promising refinement pattern.

The cell-selection overlap was 67.95% for $1N$ and 64.91% for $2N$. Thus, the observed differences arose from a substantial but not complete change in the spatial allocation of the refinement budget.

The refinement-utility analysis further showed that SOM-interface does not uniformly outperform Gradient in every spatial subset. Gradient retained lower mean post-refinement error in the common and Gradient-only regions, while SOM-interface showed lower error in the SOM-only region for the $2N$ case and in the large region selected by neither method. The global improvement is therefore interpreted as a property of the coupled CFD solution rather than as evidence that every SOM-selected cell has higher individual utility.

The appropriate conclusion at this stage is that the SOM-interface hypothesis has passed an initial controlled test and merits further investigation, but requires validation across additional flow conditions, geometries, and independent experiments before any general claim can be made.

9.3 Future Work

The results suggest that further development should focus on *refinement utility* rather than on further tuning of quantization error or the Hybrid-v1 coefficient against the existing M160 reference.

The first priority is therefore generalization.

The complete experiment should be repeated at additional Reynolds numbers while preserving the methodology and indicator definitions used in the present study. This will determine whether the observed behavior is specific to the $Re = 100$ flow field or persists as the cavity dynamics change.

A second priority is evaluation on a different CFD problem. A useful test should introduce different flow structures while remaining simple enough to permit systematic mesh verification and controlled reference solutions. This would provide a stronger test of whether SOM-derived regime information transfers beyond the lid-driven cavity.

Future work should also examine whether refinement utility can be estimated more directly. Instead of asking whether a cell has a large gradient, a large quantization error, or lies near a SOM regime transition, the relevant question is whether refining that cell or region produces a measurable improvement in a specified quantity of interest.

Conceptually, this corresponds to learning or estimating a mapping of the form

$$(\text{local flow state, numerical indicators, mesh state}) \longrightarrow \text{expected refinement benefit.} \quad (58)$$

Such a formulation would require carefully designed training data and strict separation between model development and final validation.

Additional investigations should include:

- repeating the experiment for additional Reynolds numbers;
- testing the methodology on at least one different benchmark geometry or flow configuration;
- separating spatial and temporal discretization effects through independent time-step refinement;
- evaluating quantities of interest in addition to global velocity-field discrepancies;
- investigating repeated adaptation cycles rather than only one-shot refinement;
- examining alternative SOM feature sets and map sizes through controlled ablation studies;
- comparing SOM-derived information with established error-estimation or adaptation methods;
- measuring total adaptive-workflow cost, including pilot simulation, feature extraction, machine-learning operations, remeshing, and final CFD solution;
- repeating timing measurements before drawing conclusions about computational efficiency.

Importantly, future indicator development should avoid optimizing parameters against the same fine-grid reference subsequently used for evaluation. New hypotheses should be defined using pilot-solution information and then tested against independent numerical references.

The longer-term objective is therefore not to replace conventional CFD adaptation with a SOM. It is to determine whether learned representations of the physical flow contain information that can complement established numerical indicators and improve the allocation of computational resources in a reproducible and generalizable manner.